

Requirements Table

Academic Elective Bidding & Allocation System — Problem Statement #05

Actors: Student, Academic Registrar, Notification Service

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.	High	Pass: Credits deduct accurately and prerequisites validated. Fail: Total allocated credits exceed 100 or unmet prerequisites bypassed.	Bidding credits are the core mechanism for expressing relative preference strength; without a cap and prerequisite check, students could game the system or enroll in ineligible electives.
FR-002	Functional	The system shall allow students to rank between 3 and 8 elective preferences in strict order before submitting a bid, and shall allow an existing bid to be edited until the bidding window closes.	High	Pass: Bid is saved with a unique rank per preference and remains editable until the deadline. Fail: Duplicate ranks are accepted, or a bid is edited after the window closes.	Ranked order lets the solver break ties beyond raw credits, and an editable window mirrors real bidding systems where students refine choices as information changes.
FR-003	Functional	The system shall run an allocation solver that assigns each student to at most one section per elective and shall guarantee that no student receives two electives with overlapping timetable slots.	High	Pass: Generated allocation contains zero double-bookings and zero seat-capacity violations. Fail: Any student is assigned overlapping sections or an elective exceeds its seat cap.	Timetable-collision prevention is the central value of the system; an allocation a student cannot physically attend is a functional failure.
FR-004	Functional	The system shall notify each student of their final allocation result (allocated elective, waitlist, or rejection reason) in-app and via email within 1 hour of the Registrar publishing results.	Medium	Pass: 100% of students receive an in-app and email notification with the correct outcome within 1 hour. Fail: Any student is missing a notification or receives an incorrect outcome.	Students need timely, unambiguous confirmation to plan drop/add windows and their semester; silent or delayed results erode trust in the automated process.
FR-005	Functional	The system shall allow a student to submit one allocation appeal within 48 hours of results publication, and shall allow the Academic Registrar to approve, reject, or manually reassign the appeal with a recorded justification.	Medium	Pass: Appeal is timestamped, linked to the original bid, and the decision plus justification is stored and visible to the student. Fail: An appeal is accepted after 48 hours, or a decision has no justification text.	Automated solvers can make edge-case errors (e.g., a manually waived prerequisite); a bounded, auditable appeal path lets the Registrar correct errors without reopening the whole round.
NFR-001	Nonfunctional	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	The bidding window is short and university-wide, so the solver must complete well within the published results deadline.
NFR-002	Nonfunctional	The bidding portal shall maintain 99.5% uptime during the published bidding window and remain fully usable on both desktop and mobile browsers without a dedicated app.	High	Pass: Uptime monitoring shows $\geq 99.5\%$ availability during the bidding window, and all core actions complete on a mobile viewport. Fail: Downtime exceeds the 0.5% error budget, or any core action is unusable on mobile.	Bidding happens in a fixed, non-negotiable window shared by thousands of students; an outage or unusable mobile UI unfairly disadvantages students without desktop access.